

RESEARCH STUDY & ACADEMIC DATA REPORT

Synthetic multi-study report for testing PDF search, extraction, table recognition, citations, statistical values, formulas, and long-form mixed content.

Study Program	Learning Analytics 2026	Participants	2,480
Institutions	18	Survey Responses	2,146
Completion Rate	86.53%	Median Score	74.2 / 100
Confidence Level	95%	Data Quality	96.8% complete

1. Study Overview

The synthetic study evaluates how different instructional interventions affect learner engagement, assessment outcomes, retention, and task completion. The dataset combines cohort-level results, survey responses, experiment arms, and statistical summaries.

2. Experimental Cohorts

Cohort	Intervention	N	Pre-Test Mean	Post-Test Mean	Gain	Retention	p-value	Decision
C-01	Baseline curriculum	232	55.8	73.4	+17.5	77.3%	0.021	Significant
C-02	Adaptive practice	136	57.2	62.3	+5.0	86.7%	0.001	Significant
C-03	Baseline curriculum	158	66.7	83.9	+17.3	73.1%	0.004	Significant
C-04	Project-based learning	262	65.6	78.1	+12.5	90.8%	0.004	Significant
C-05	Peer instruction	217	58.4	68.6	+10.2	92.5%	0.021	Significant
C-06	Spaced repetition	265	62.3	74.8	+12.5	89.5%	0.004	Significant
C-07	Baseline curriculum	174	54.1	59.2	+5.1	77.0%	0.047	Significant
C-08	Baseline curriculum	160	48.1	58.4	+10.3	85.8%	0.012	Significant
C-09	Spaced repetition	179	54.6	62.6	+8.0	92.5%	0.001	Significant
C-10	Adaptive practice	179	69.2	84.3	+15.0	86.6%	0.083	Not significant
C-11	Adaptive practice	228	50.6	57.8	+7.1	74.7%	0.012	Significant
C-12	Peer instruction	138	49.4	64.5	+15.1	85.9%	0.047	Significant

3. Survey Response Register

Response ID	Age	Program	Hours/Wk	Satisfaction	Difficulty	Recommendation	Comment Code
R-202600001	32	Computer Science	11.6	Neutral	Hard	Maybe	C08
R-202600002	32	Computer Science	14.2	Low	Moderate	No	C05
R-202600003	18	Computer Science	7.0	Very Low	Very Hard	Yes	C05
R-202600004	36	Computer Science	17.4	High	Hard	No	C08
R-202600005	26	Business Analytics	21.9	Low	Hard	Yes	C08
R-202600006	29	Design	9.0	Very High	Easy	Maybe	C07
R-202600007	40	Design	8.1	Low	Very Hard	Yes	C08
R-202600008	20	Mechanical Engineering	7.4	Very High	Hard	No	C01
R-202600009	39	Mathematics	11.5	Very High	Easy	No	C06
R-202600010	37	Design	14.7	Very High	Easy	Maybe	C03
R-202600011	27	Design	19.7	Low	Very Easy	Yes	C04
R-202600012	40	Business Analytics	8.6	Very High	Very Hard	Maybe	C02
R-202600013	25	Computer Science	11.5	Very High	Easy	Maybe	C05

Response ID	Age	Program	Hours/Wk	Satisfaction	Difficulty	Recommendation	Comment Code
R-202600014	34	Information Systems	7.8	Low	Hard	Yes	C08
R-202600015	35	Business Analytics	10.0	Very Low	Very Hard	No	C03
R-202600016	20	Mechanical Engineering	17.6	Low	Moderate	No	C05
R-202600017	18	Mechanical Engineering	16.7	Very Low	Very Hard	No	C08
R-202600018	22	Design	13.0	Neutral	Moderate	No	C07
R-202600019	32	Mechanical Engineering	19.5	High	Very Hard	Maybe	C04
R-202600020	42	Design	3.4	Low	Hard	Maybe	C03
R-202600021	33	Computer Science	5.0	Very Low	Moderate	Yes	C08
R-202600022	21	Mathematics	20.3	Very High	Easy	Maybe	C03
R-202600023	20	Design	17.7	Neutral	Moderate	No	C07
R-202600024	38	Computer Science	19.1	Very Low	Hard	Maybe	C08
R-202600025	35	Computer Science	14.5	High	Moderate	Yes	C02
R-202600026	31	Computer Science	17.3	Very High	Hard	Yes	C05
R-202600027	18	Computer Science	8.8	Very High	Moderate	Maybe	C06
R-202600028	31	Business Analytics	7.3	Low	Very Hard	No	C03
R-202600029	23	Business Analytics	4.0	Low	Very Hard	No	C04
R-202600030	33	Mathematics	5.3	Low	Moderate	Maybe	C05
R-202600031	39	Computer Science	20.0	Low	Moderate	No	C03
R-202600032	20	Design	20.9	Very Low	Moderate	No	C07
R-202600033	40	Mechanical Engineering	11.4	Neutral	Easy	Maybe	C08
R-202600034	21	Business Analytics	9.9	Neutral	Very Hard	Maybe	C05
R-202600035	18	Information Systems	10.2	Very High	Very Hard	No	C01
R-202600036	37	Information Systems	12.2	Neutral	Easy	No	C06
R-202600037	25	Information Systems	6.2	Neutral	Easy	No	C02
R-202600038	38	Information Systems	3.3	Low	Very Easy	No	C06
R-202600039	41	Business Analytics	4.3	Neutral	Moderate	No	C07
R-202600040	23	Business Analytics	5.1	Very Low	Moderate	No	C05
R-202600041	23	Mechanical Engineering	20.3	Low	Moderate	No	C06
R-202600042	21	Design	21.3	High	Easy	No	C07
R-202600043	35	Mechanical Engineering	4.3	Low	Very Easy	Maybe	C02
R-202600044	32	Mechanical Engineering	15.6	Neutral	Very Hard	Maybe	C06
R-202600045	21	Information Systems	7.1	Very High	Very Hard	No	C06
R-202600046	37	Business Analytics	15.1	Low	Moderate	No	C07
R-202600047	21	Business Analytics	3.4	Very High	Moderate	Maybe	C02
R-202600048	21	Business Analytics	19.8	High	Hard	Maybe	C06
R-202600049	39	Information Systems	16.1	Very Low	Hard	No	C03
R-202600050	31	Information Systems	4.4	Low	Very Hard	Maybe	C05

4. Statistical Summary

Metric	Baseline	Intervention	Difference	95% CI	Effect Size	Interpretation
Assessment score	68.4	74.2	+5.8	[4.9, 6.7]	0.41	Moderate positive effect
Weekly engagement	6.8 h	8.1 h	+1.3 h	[0.9, 1.7]	0.36	Small-to-moderate effect
Course completion	78.2%	86.5%	+8.3 pp	[5.2, 11.4]	0.52	Meaningful improvement
Retention after 90d	71.4%	82.7%	+11.3 pp	[7.6, 15.0]	0.61	Strong improvement
Reported workload	5.2	5.8	+0.6	[0.2, 1.0]	0.18	Small increase

5. Methodology & Data Dictionary

- Sampling:** Stratified synthetic sampling across 18 institutions and six academic program families.
- Primary outcome:** Post-test assessment score on a normalized 0–100 scale.
- Retention:** Percentage of participants completing the follow-up assessment approximately 90 days after intervention.
- Effect size:** Standardized mean difference; values above 0.50 are interpreted here as comparatively strong.
- Missingness:** Records with missing primary outcomes are retained and marked rather than silently removed.
- Privacy:** All names, identifiers, comments, and measurements in this document are synthetic and do not represent real participants.

6. Sample Reference Notes

Ref.	Source Type	Citation Token	Topic	Year
[R1]	Journal Article	DOI:10.0000/SYN-2026-001	Adaptive learning	2026
[R2]	Conference Paper	CONF-EDU-26-117	Peer instruction	2026
[R3]	Technical Report	TR-LA-2026-09	Learning analytics	2026
[R4]	Dataset Card	DATA-SYN-2604	Synthetic cohort dataset	2026

7. Quality-Control Checklist

- All cohort totals reconcile to the participant summary.
- Percentages are retained to at least one decimal place where applicable.
- Identifiers remain unique within their respective tables.
- Statistical values preserve signs, decimal points, brackets, and units.
- Synthetic-data disclaimer is present and must not be interpreted as a real study result.